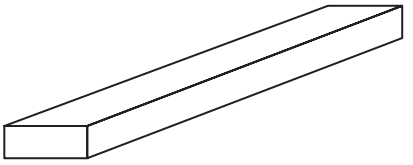
22 A solid cuboid has a volume of 40 cm^3
The cuboid has a total surface area of 100 cm^2
One edge of the cuboid has length 2 cm.

Find the length of a diagonal of the cuboid.
Give your answer correct to 3 significant figures.



Let the other edges be xcm and ycm.

Volume: $2xy = 40 \rightarrow xy = 20$.

1 mark
Volume equation

Surface area: $2(xy + 2x + 2y) = 100$.

1 mark
All six faces

$40 + 4x + 4y = 100 \rightarrow x + y = 15$.

$y = 20/x$: $x + 20/x = 15$.

1 mark
Eliminate y

$x^2 - 15x + 20 = 0$.

$x = (15 \pm \sqrt{145}) \div 2$.

1 mark
Solve quadratic

Edges x,y: 1.4792027... and 13.5207973... cm.

Diagonal$^2 = 2^2 + x^2 + y^2$.

1 mark
3D Pythagoras

$= 4 + (x + y)^2 - 2xy = 4 + 225 - 40 = 189$.

Diagonal $= \sqrt{189} = 13.747727... \approx 13.7$cm (3sf).

1 mark
13.7cm

.....cm

(Total for Question 22 is 6 marks)